Lab – Digital Forensics Analysis Using Autopsy

Overview

In a previous lab, we installed Autopsy onto a standalone virtual Windows 10 Pro machine. In this lab, we will see how we create a case in Autopsy, import a forensic image, and conduct a forensic analysis of a suspect's computer.

Lab Requirements

One Windows 10 Pro machine (Virtual or otherwise)

One install of Autopsy

One copy of the forensic image file, 4Dell Latitude CPi.E01

Lab Scenario

It is suspected that this computer was used for hacking purposes, although it cannot be tied to a hacking suspect, Greg Schardt. Schardt also goes by the online nickname of "Mr. Evil," and some of his associates have said that he would park his vehicle within range of Wireless Access Points where he would then intercept internet traffic, attempting to get credit card numbers, usernames & passwords. Find any hacking software, evidence of their use, and any data that might have been generated. Attempt to tie the computer to the suspect, Greg Schardt.

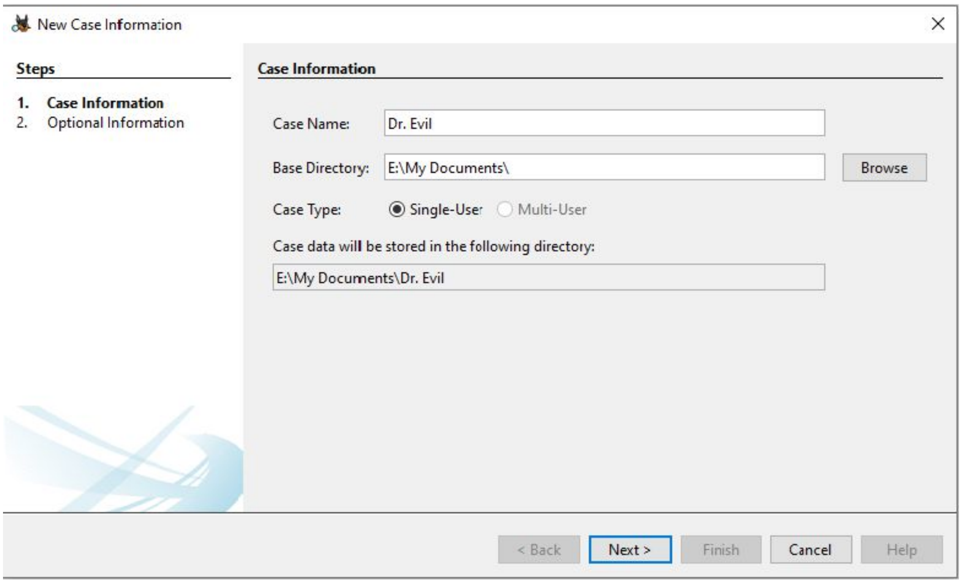
Begin the Lab!

Launch Autopsy either for your virtual installation of Windows 10 or your personal device.

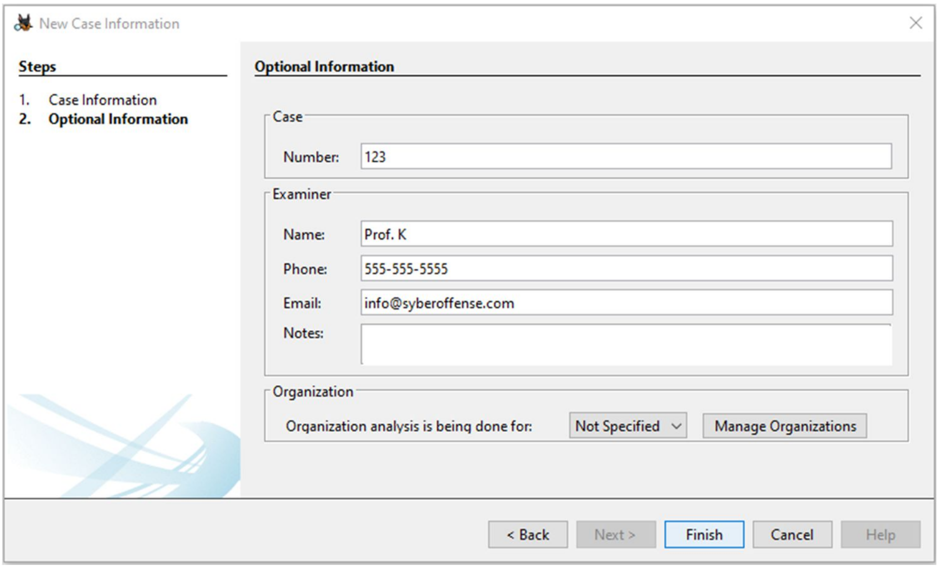
Once Autopsy loads, click the option to create a new case.

Provide the Case Name and the directory to store the case file. Click on Next.

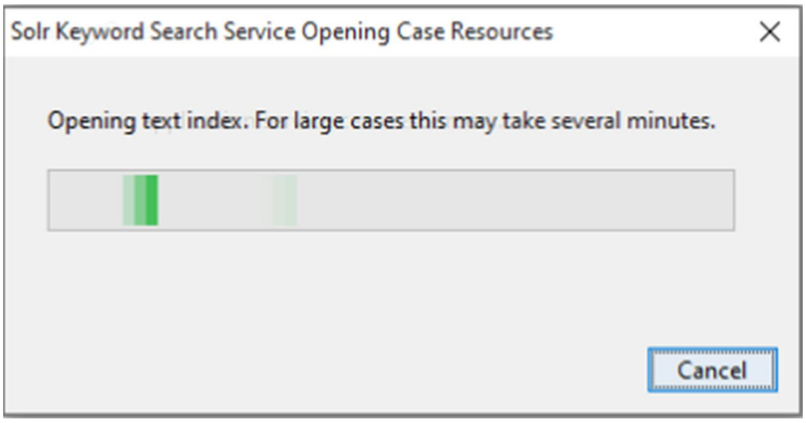
You are free to name your case as you see fit.



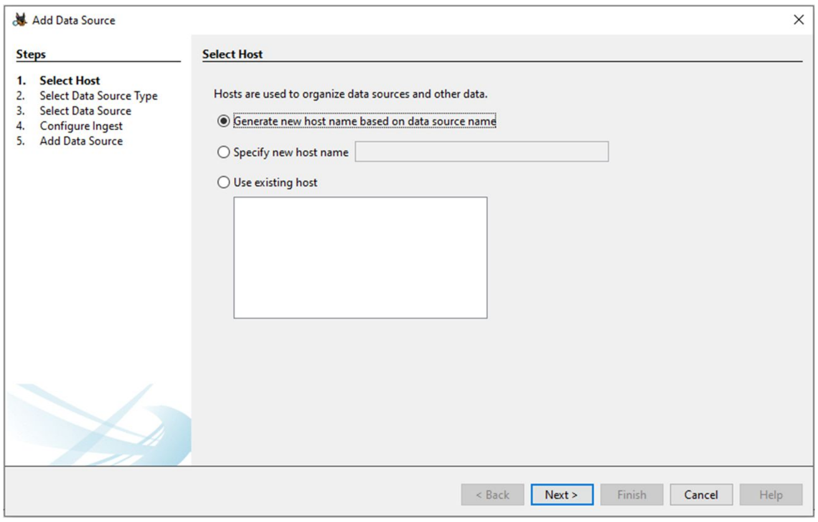
Add Case Number and Examiner’s details, then click on Finish.



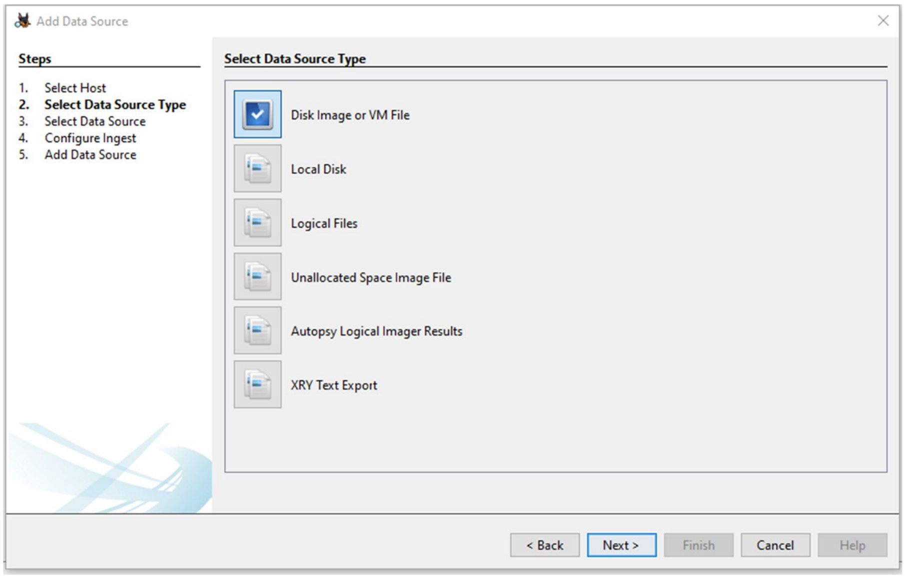
Autopsy generates a case file.



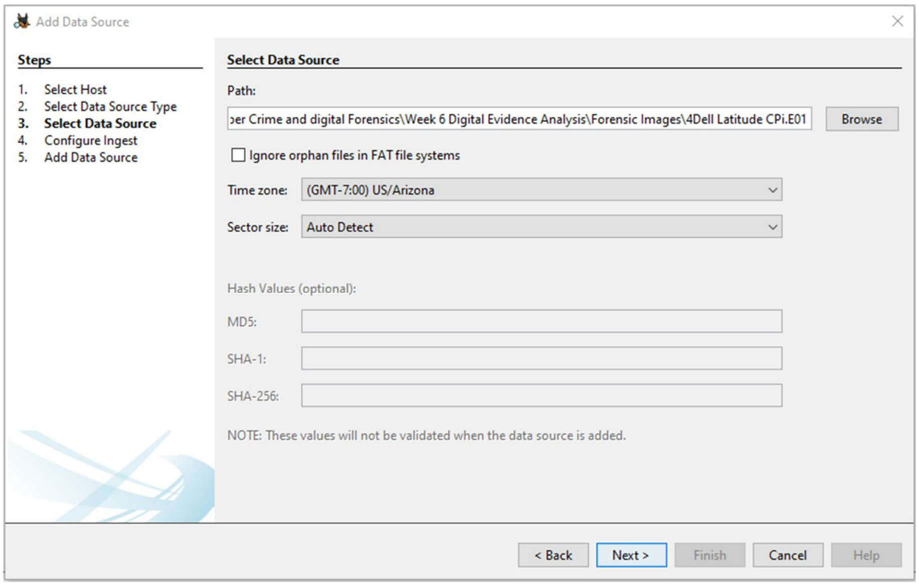
On the Add Data Source page, accept the defaults and click next.



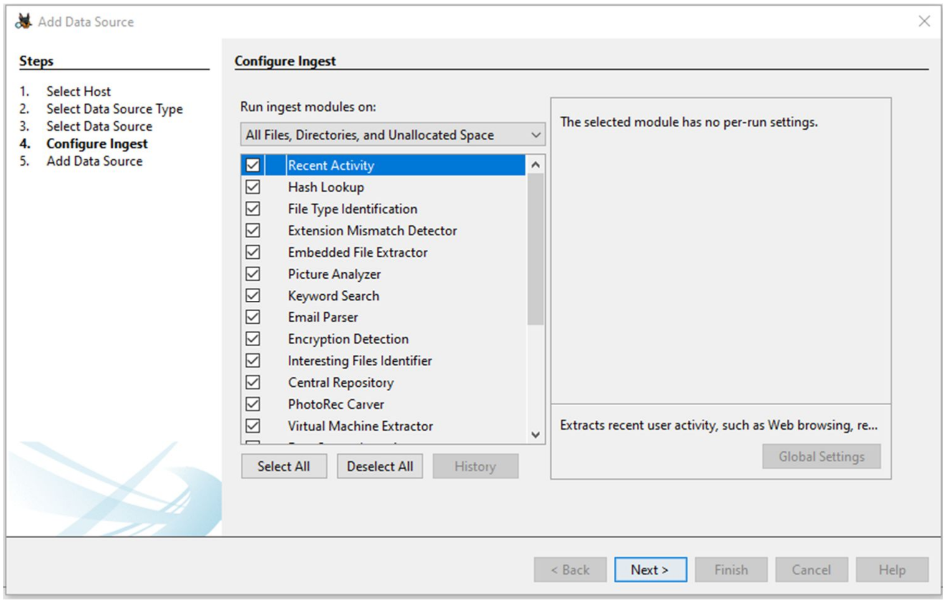
Choose the required data source type, in this case, Disk Image, and click on Next.



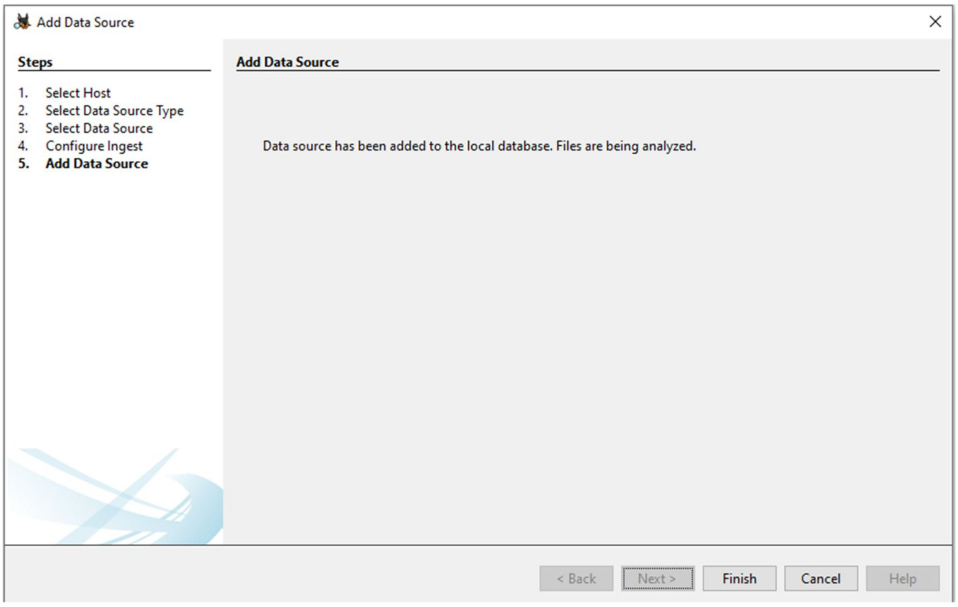
Give path of the data source and click on *Next.*



Accept the default modules that are checked and click Next.



After the data source has been added, click on Finish.



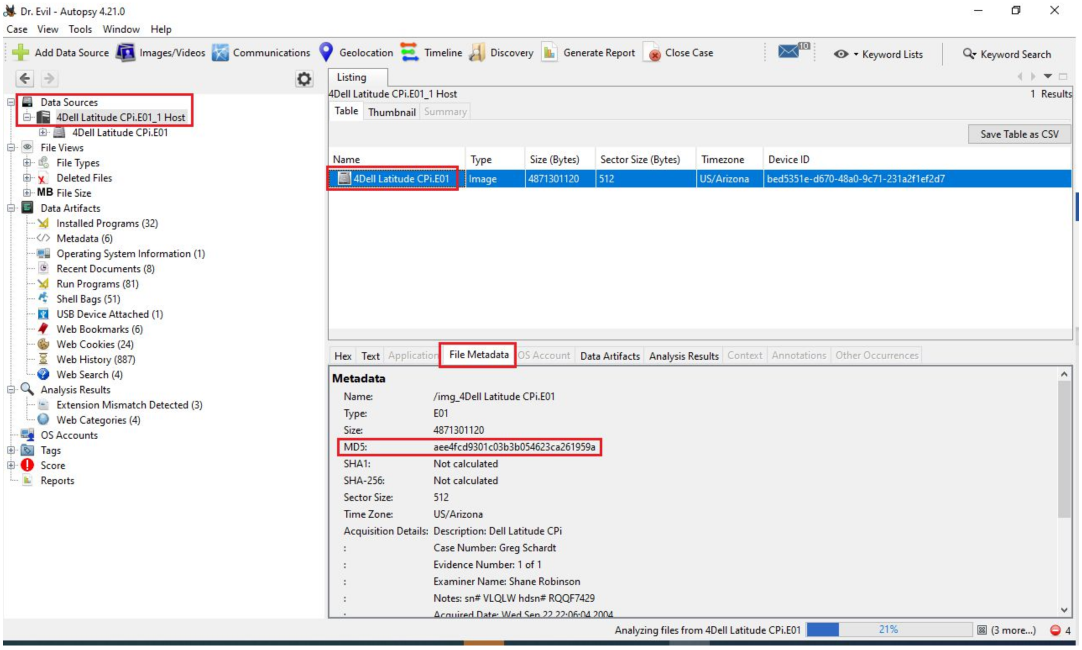
Once all the modules have been ingested, you can begin investigating. It is recommended waiting until the analysis and integrity check are complete.

Begin your analysis by answering the following questions.

Q1. What is the image hash?

Answer: AEE4FCD9301C03B3B054623CA261959A.

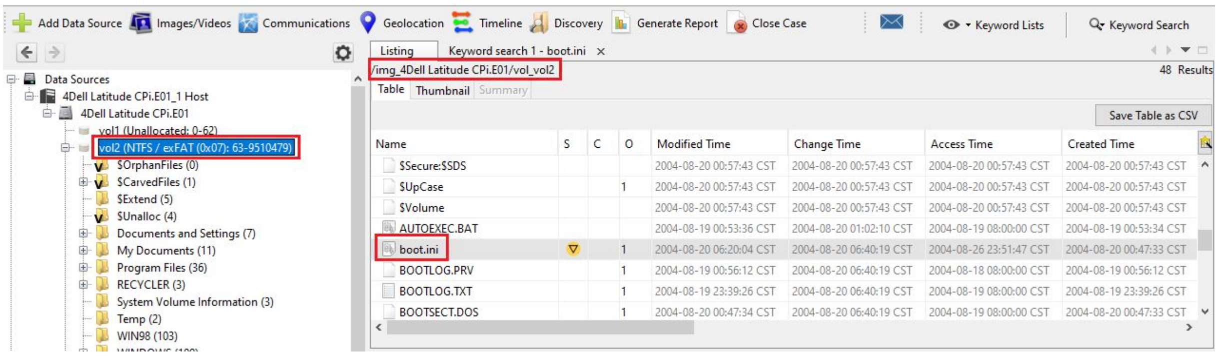
To check the image hash, click on the image and go to the File Metadata tab. (We check the image hash to verify that it is the same as the hash created during the time when the image was created.)

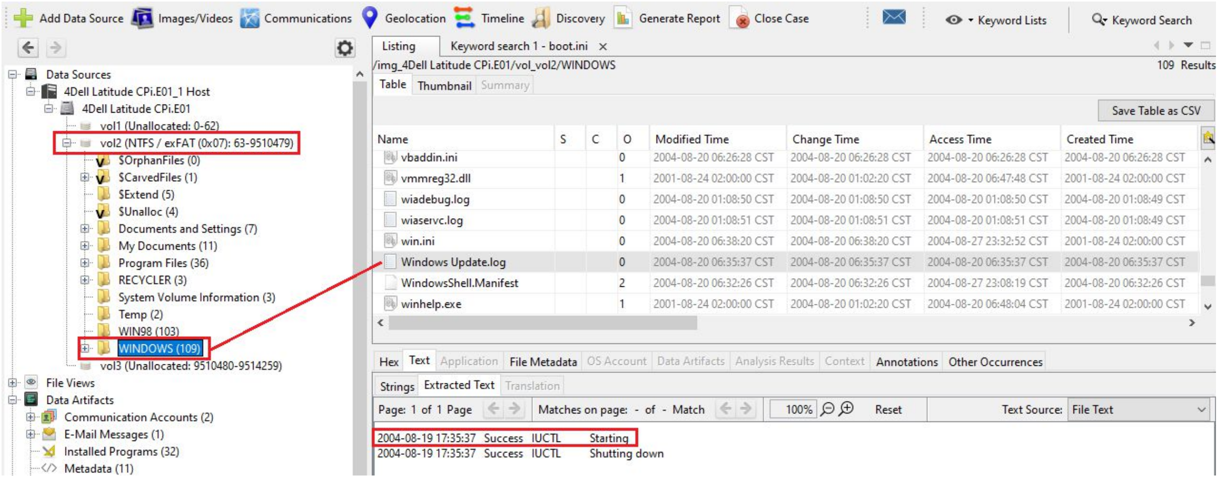


Q2: What operating system was used on the computer?

Answer:

Look in the C:\boot.ini file.

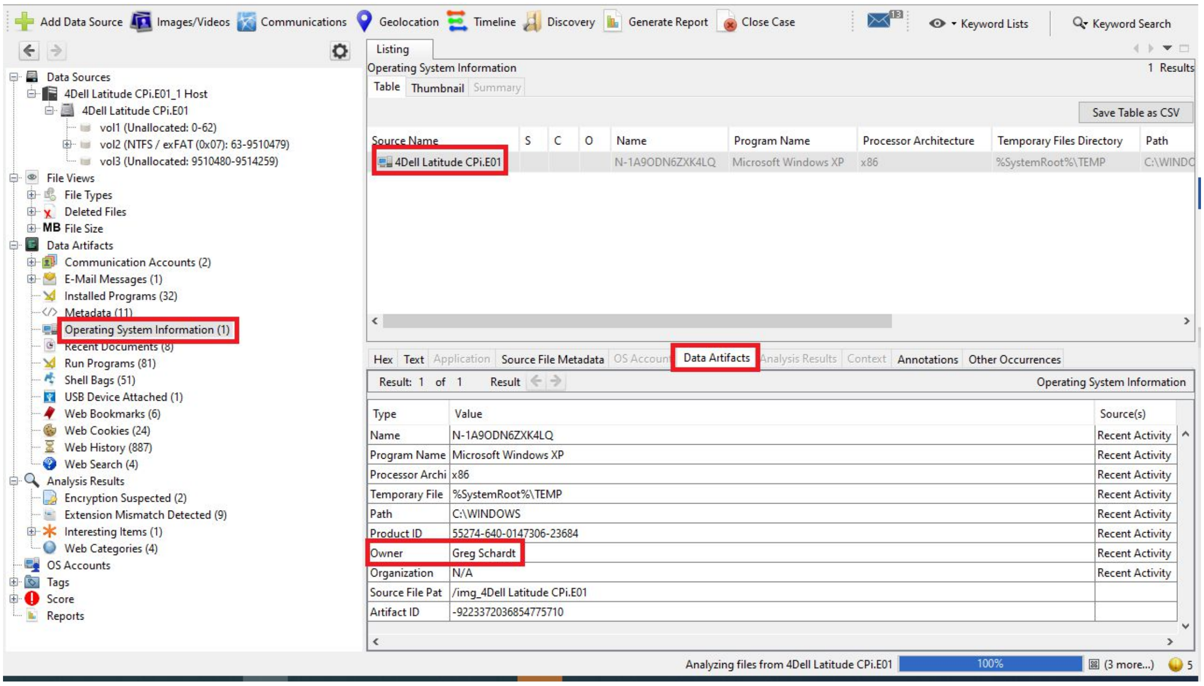




Q4. Who is the registered owner?

Answer. Greg Schardt

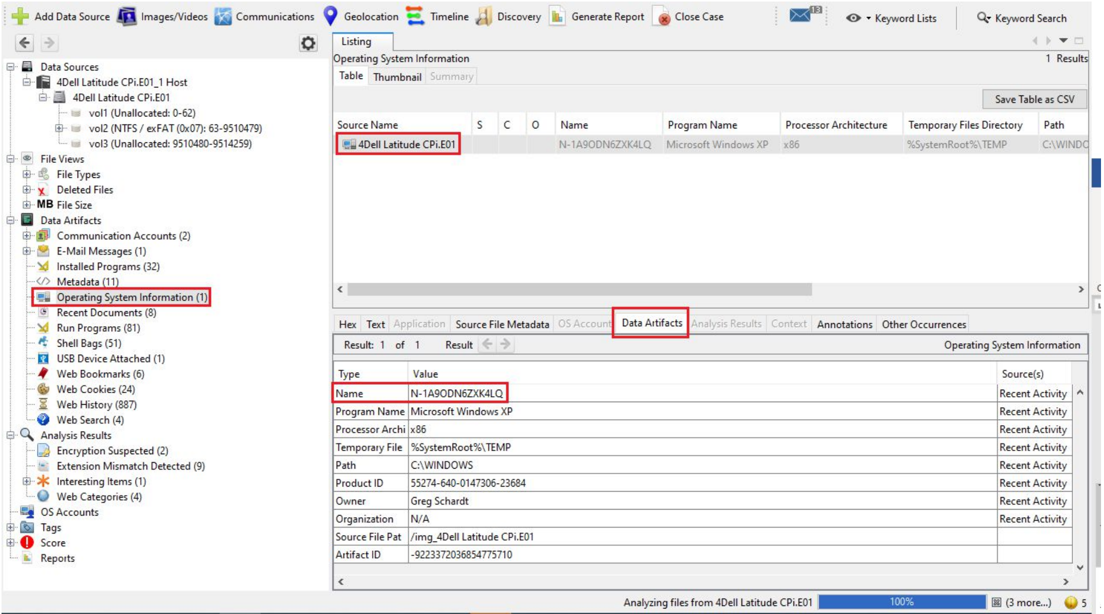
Answer. (Click on *Operating System Information*)



Q5. What is the computer account name?

Answer. The account name is N-1A9ODN6ZXK4LQ

(Click on *Operating System Information*)

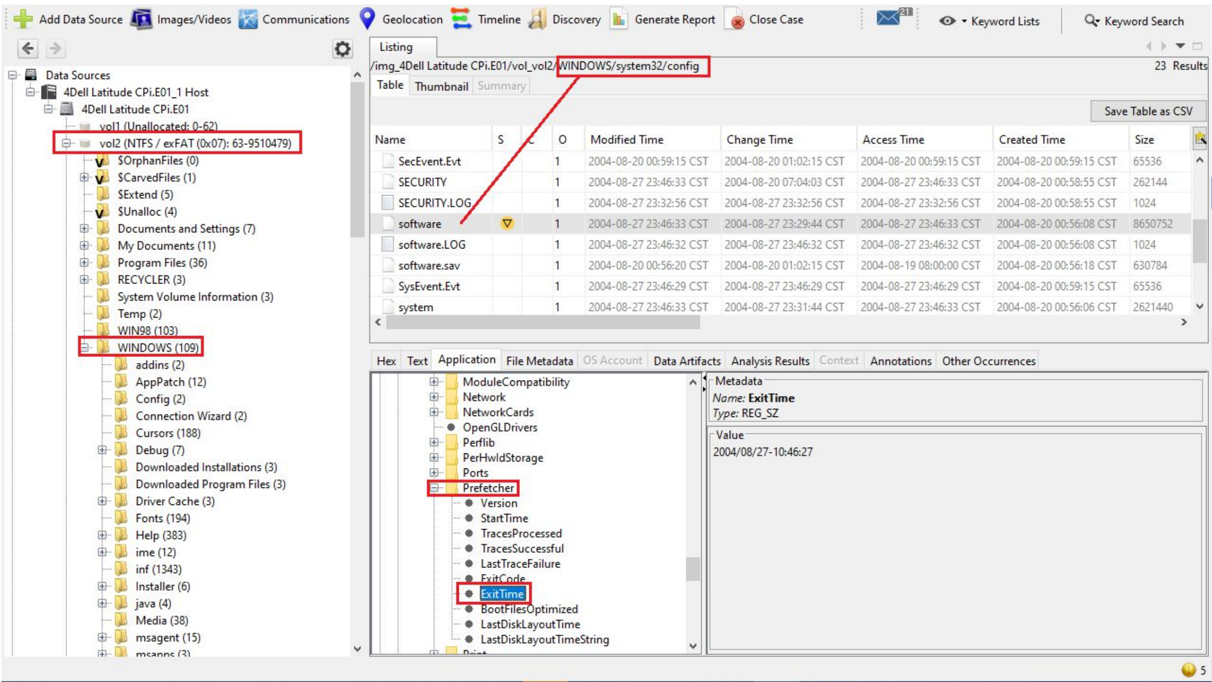


Q6. When was the last recorded computer shutdown date/time?

Answer: The last recorded shutdown time of the computer is 2004/08/27-10:46:27

To find this, go to:

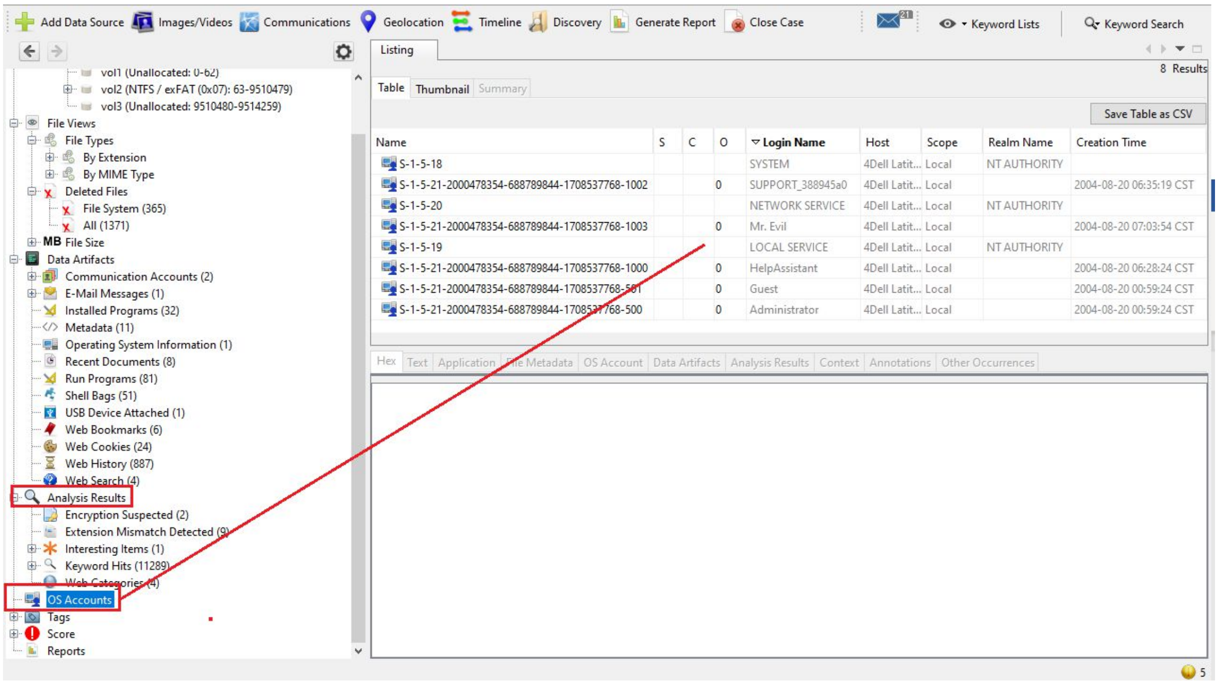
C:\WINDOWS\system32\config\software\Microsoft\WindowNT\CurrentVersion\Prefetcher\ExitTime



Q7. How many accounts are recorded (total number)?

Answer: 5 accounts, Mr, Evil, Administrator, Guest, Support388945a0, and HelpAssistant

In the left side panel, we go to Analysis Results > OS Account



Q8 .Who was the last user to logon to the computer?

Answer:: Mr. Evil

The system will obtain the last user who logged on from the key ‘DefaultUserName’. This information can be uncovered from the following path

Click on Data Sources select 4Dell Latitude --> vol2 -->

WINDOWS/SYSTEM32/CONFIG/SOFTWARE/MICROSOFT/WINDOWS NT/CURRENT

VERSION/WINLOGON/DEFAULT USER NAME

Q9 . List the network cards used by this computer.

Answer:

- Compaq WL110 Wireless LAN PC Card,

- Xircom CardBus Ethernet 100 + Modem 56 (Ethernet Interface).

Click on Data Sources select 4Dell Latitude --> vol2 -->

WINDOWS\system32\config\software\Microsoft\Windows NT\CurrentVersion\NetworkCards\

click on descriptions.

Q10 . What is the IP address and MAC address of the computer?

Answer: IP=192.168.1.111, MAC=00:10:a4:93:3e:09

Click on Data Sources select 4Dell Latitude --> vol2 --> Program Files/Look@LAN/irunin.ini

Q11 . Search for programs/tools that aided in the crime (Wireless Hacking)

Ans11: The programs which will be used for hacking purpose

1. Look@LAN - An advanced network monitor that allows you to monitor your net in a few clicks.

2. Cain and Abel – A password recovery tool for Microsoft Windows. It can recover many kinds of passwords using methods such as network packet sniffing and cracking various password hashes by using methods such as dictionary attacks, brute force, and cryptanalysis attacks.

3. Network Stumbler - A tool for Windows that facilitates the detection of Wireless LANs using the 802.11b, 802.11a, and 802.11g WLAN standards. It runs on Microsoft Windows operating systems from Windows 2000 to Windows XP.

4. mIRC- An Internet Relay Chat client for Windows, created in 1995.

5. Ethereal/Wireshark - A free and open-source packet analyzer. It is used for network troubleshooting, analysis, software, and communications protocol development, and education.

6. 123WASP - WASP will display all passwords of the currently logged-in user that are stored in the Microsoft PWL file.

Click on Results --> Extracted Content --> Installed Programs

Q12 . Which Email client is used by Mr. Evil?

Answer: Outlook Express, Forte Agent, MSN Explorer, MSN (Hotmail) Email

Click on Data Sources select 4Dell Latitude --> vol2 -->

WINDOWS\system32\config\software\clients\Mail

Q13 . What is the SMTP email address for Mr. Evil?

Answer: iswhoknowsme@sbcglobal.net

Click on Data Sources select 4Dell Latitude --> vol2 --> Program Files\Agent\Data\Agent.ini

Q14 . How many executable files are in the recycle bin?

Answer: There are 4 files in the recycle bin.

Click on Data Sources select 4Dell Latitude --> vol2 -->Recycler

Q15 . Is there any malware on the computer?

Answer: There is zip bomb malware by the name of unix_hack.giz in this system.

Click on Results --> Extracted Content --> Interesting Items --> Possible Zip Bomb

-->Interesting Files

Q16 . A popular IRC (Internet Relay Chat) program called MIRC was installed. What is the userid?

Answer: user=Mini Me, email=none@of.ya, nick=Mr, anick=mrevilrulez

How ?

Click on Data Sources select 4Dell Latitude --> vol2 -->Program Files\mIRC\mirc.ini

Q17 . Ethereal, a popular “sniffing” program that can be used to intercept wired and wireless internet packets was also found to be installed. When TCP packets are collected and re-assembled, the default save directory is the user’s \My Documents directory. What is the name of the file that contains the intercepted data?

Answer: The file name is Interception

How?

Click on Data Sources select 4Dell Latitude --> vol2 --> Document and

Settings\Mr.Evil\intercerption

Q18 . Which internet browser was used?

Answer: Internet Explorer

Click on Data Sources select 4Dell Latitude --> vol2 --> Document and

Settings\Mr.Evil\intercerption

scroll down and see User-Agent: Mozilla/4.0 (compatible; MSIE 4.01; Windows CE; PPC;240x320)

End of the lab